Xinrui Dong

M.Sc. Student in Probability and Mathematical Statistics
sinzhui.dun@math.msu.ru | +86 186 5388 0441 | Moscow, Russia
Academic homepage: xinrui-dong.github.io

Research Interests

- Branching processes and catalytic branching random walks; stochastic processes; probability theory
- Markov processes, limit theorems and stochastic analysis; ergodic methods in mathematical finance

Education

Lomonosov Moscow State University (MSU) Sep 2025 - Jun 2027 (expected)
M.Sc. in Mathematics, Probability and Mathematical Statistics Moscow, Russia

- Faculty of Mechanics and Mathematics; Department of Mathematical Statistics and Stochastic Processes
- Academic advisor: Associate Professor Ekaterina V. Bulinskaya
- Current research area: catalytic branching random walks; developing an explicit propagation-front project as the prospective basis of the master's thesis
- Theoretical preparation: real analysis, measure theory, martingales, renewal theory, large deviations and multitype branching processes

Lomonosov Moscow State University, Institute of Russian Language and Culture Sep 2024 - Jun 2025
Preparatory Programme in Russian, Economics Track Moscow, Russia

- Completed an intensive one-year Russian programme and received an MSU Russian Language Proficiency Certificate
- Supplementary mathematical study included Mathematical Analysis and Financial Mathematics

Xiamen University Tan Kah Kee College Sep 2020 - Jul 2024
Bachelor of Engineering (B.Eng.) Zhangzhou, China

- Weighted average: 81.5/100; rank: 28/77
- Selected coursework: Linear Algebra, Probability and Statistics, and Mathematical Modelling

Selected Research Projects

Course Paper: From the Galton-Watson Process to Catalytic Branching Random Walks 2026
Expository review; grade: 5/5; supervised by Associate Professor Ekaterina V. Bulinskaya

- Developed a structured review linking the Galton-Watson process to spatially inhomogeneous branching models on $\mathbb{Z}^d$
- Synthesised catalyst-dependent dynamics, Perron-root criticality and propagation-front results under Cramér's condition
- Current follow-on work: constructing explicit motion kernels and offspring means for spherical, ellipsoidal and polyhedral candidate fronts, with formulas and Python visualisations

Long-Term Asset Allocation through Ergodic Theory Oct 2025 - Feb 2026

- Independently modelled a continuous-time stochastic market and self-financing portfolio using time-homogeneous Markov dynamics
- Applied Foster-Lyapunov and ergodic arguments to study invariant measures, time-average convergence and long-run portfolio growth
- Showed that asymptotic allocation and average capital growth are determined by stationary market characteristics; the resulting article was published in 2026

Publication

- **Dong, X.** (2026). Study of a Long-Term Asset Allocation Model from an Ergodic Theory Perspective. *Innovation & Investment*, no. 3. Sole-authored; published in a peer-reviewed journal included in the Russian Higher Attestation Commission (VAK) list. [In Russian]

Honours and Awards

- **First Prize**, 2nd National University Students' Mathematics Ability Challenge Jan 2025
- MSU Russian Language Proficiency Certificate Jun 2025

Technical and Language Skills

- **Methods:** branching and Markov processes, stochastic differential equations, ergodic theory, proof reconstruction, regression and Monte Carlo methods
- **Programming and tools:** Python (NumPy, Pandas, Matplotlib; statistical analysis and developing stochastic simulation), LaTeX and Excel
- **Languages:** Chinese (native); Russian (MSU certificate); English (academic reading)